

Surgical Robots

A Concise Reference for Exam Preparation

Based on the course held by Professor Vendittelli
Sapienza University of Rome

Contents

1	Historical and Pioneering Systems	4
1.1	PUMA 560	4
1.2	ROBODOC / TSolution One	4
1.3	CASPAR	4
2	MIS and Teleoperated Systems	6
2.1	ZEUS	6
2.2	da Vinci (Intuitive Surgical)	6
2.3	AESOP	7
3	Orthopaedic Systems	8
3.1	MAKO (Stryker)	8
3.2	Mazor Renaissance / MARS	8
4	Neurosurgical Systems	9
4.1	NeuroMate	9
4.2	CyberKnife	9
5	Synergistic and Passive Navigation Systems	10
5.1	PADyC	10
5.2	Viewing Wand (ISG Technologies)	10
6	Surface Tracking, Radiology and Ultrasound	11
6.1	SCALPP	11
6.2	OTELLO / MELODY	11
6.3	INNOMOTION	11
6.4	XACT Robotics	12
7	Cardiac and Microscopy Systems	13
7.1	Cardiolock	13
7.2	Pentagon Mechanism	14
8	Ophthalmic Robotic Systems	15
8.1	Steady-Hand Robot (SHR)	15
8.2	Micron (Carnegie Mellon University)	15
8.3	Leuven Robot	15
8.4	TU Munich Ophthalmic Robot	16
9	Radiotherapy and Application-Specific Systems	17
9.1	ROBHOT / Alba ON4000	17
9.2	Surgiscope	17

10 Quick Reference Table

18

Chapter 1

Historical and Pioneering Systems

PUMA 560

Origin / Manufacturer: Unimation / Stäubli
Year: 1985
Domain: Neurosurgery
Troccaz / LoA: Active — LoA 2

The PUMA 560 is a standard 6-DOF anthropomorphic industrial arm notable for being the **first robot used in a surgical procedure**: a CT-guided stereotactic brain biopsy in 1985. It was an experimental repurposing of an industrial platform; no dedicated surgical safety features were present. Its significance is entirely historical: it proved the feasibility of robot-assisted intervention.

ROBODOC / TSolution One

Origin / Manufacturer: ISS / Think Surgical (now Curexo)
Year: 1992 (first clinical)
Domain: Orthopaedics — Total Hip/Knee Replacement
Troccaz / LoA: Active — LoA 2 (Task-level autonomy)

ROBODOC was the **first robot designed specifically for autonomous surgical bone milling**. Kinematic architecture: **SCARA** arm with 2-DOF wrist and **6-DOF force/torque sensor** at the end-effector. Bone-milling accuracy: < 1 mm. Pre-operative planning via the companion **Orthodoc** software on CT data.

Registration used **point-to-point (Procrustes)** on titanium fiducial pins implanted in the femur via a **separate minor surgical procedure under anaesthesia**, days before the main operation. This invasiveness was the primary **workflow burden** limiting clinical adoption. The system evolved into the **TSolution One**.

CASPAR

Origin / Manufacturer: URS Ortho, Germany
Year: Late 1990s
Domain: Orthopaedics
Troccaz / LoA: Active — LoA 2

Functionally analogous to ROBODOC but using an **anthropomorphic** rather than SCARA arm. Its engineering contribution over ROBODOC is a **second fiducial set S'** tracked contin-

uously intraoperatively, compensating for patient movement during milling — a problem ROBODOC could not handle without full patient immobilisation.

Chapter 2

MIS and Teleoperated Systems

ZEUS

Origin / Manufacturer: Computer Motion (acquired by Intuitive Surgical, 2003)

Year: Late 1990s

Domain: MIS / Telesurgery

Troccaz / LoA: Teleoperated — LoA 1

ZEUS was the direct competitor of the da Vinci and the system used in **Operation Lindbergh** (2001): the first transatlantic telesurgery, a laparoscopic cholecystectomy performed from New York on a patient in Strasbourg. Duration: **54 minutes**; round-trip ATM fibre-optic latency: $\approx$ **150 ms**. Note: the documented surgeon performance degradation threshold is **200–300 ms**; Lindbergh succeeded because latency was below this threshold.

Kinematic architecture: **PRRRR** — three actuated joints plus a **passive universal joint** at the trocar, satisfying the RCM via **passive joints**. Intrinsically safe but imprecise. Discontinued after Computer Motion's acquisition by Intuitive Surgical in 2003.

da Vinci (Intuitive Surgical)

Origin / Manufacturer: Intuitive Surgical

Year: 1999 (FDA clearance)

Domain: MIS — General, Urological, Gynaecological Surgery

Troccaz / LoA: Teleoperated — LoA 1

The da Vinci is the dominant teleoperated surgical platform. It addresses every kinematic and ergonomic limitation of manual laparoscopy:

- **7 DOF at the tool tip** via articulated **EndoWrist** instruments (trocar provides 4 DOF; the articulated wrist adds 3).
- **10× HD 3D stereoscopic vision**: restores depth perception lost in laparoscopy.
- **Motion scaling 3:1 or 5:1**: the slave moves 3–5× less than the master, filtering physiological tremor ($\sim 50\text{--}200\ \mu\text{m}$) and amplifying fine motor precision.
- **Mechanical RCM** via an articulated parallelogram linkage, modelled as a planar **RRRR four-bar**:

$$\dot{\vartheta}_c = \frac{s_2}{s_3} \dot{\vartheta}_1$$

Collapse (Type 2 singularity, loss of RCM) occurs at $s_3 = 0$ ($\vartheta_3 = \pm\pi$) — strictly forbidden.

- Patient cart: 3–4 robotic arms; surgeon operates from an ergonomic seated console.

Market data (2024): >10,000 units worldwide; one procedure every 11.75 s; unit cost \$1.0M–\$2.3M. The da Vinci is described as "*laparoscopy with superpowers*": it retains the 5–12 mm port

access of laparoscopy while restoring the dexterity and vision of open surgery.

AESOP

Origin / Manufacturer: Computer Motion

Year: Early 1990s

Domain: MIS — Endoscope Holder

Troccaz / LoA: Active / Auxiliary Surgical Support — LoA 2

AESOP (Automated Endoscopic System for Optimal Positioning) is a **7-DOF, voice-controlled** robotic arm that holds and positions the laparoscopic camera, eliminating the need for a human camera assistant and the associated fatigue-induced instability. In Taylor's CIS framework it is an **Auxiliary Surgical Support**: it works alongside the surgeon to hold an instrument rather than extending dexterity. Discontinued after acquisition by Intuitive Surgical.

Chapter 3

Orthopaedic Systems

MAKO (Stryker)

Origin / Manufacturer: Stryker — evolved from Acrobot, Imperial College London

Year: 2006/2008

Domain: Orthopaedics — Knee and Hip Arthroplasty

Troccaz / LoA: Synergistic — LoA 1

MAKO is the commercial evolution of the **Acrobot** (Active Constraint Robot). It enforces **Forbidden-Region Virtual Fixtures** via **impedance control** in a three-region spatial logic:

- **Region I** ($d > 2\text{ mm}$): free motion; low impedance.
- **Region II** ($0 < d \leq 2\text{ mm}$): stiffness increases progressively. A step function would cause overshoot instability.
- **Region III** ($d \leq 0$): maximum PD gains; robot actively pushes back.

Dual-loop control: inner joint loop at **550 Hz / 2 ms**; outer Cartesian admittance loop at **100 Hz / 10–15 ms**. End-effector range: ± 30 yaw/pitch, 30–50 cm extension. **Clinical outcomes (2003 Acrobot trial):** accuracy 2–3 $\times$ better, reproducibility 3 $\times$ better than manual surgery; patient satisfaction 92% at two years.

Mazor Renaissance / MARS

Origin / Manufacturer: Mazor Surgical Technologies (now Medtronic)

Year: 2004

Domain: Orthopaedics — Spinal Surgery

Troccaz / LoA: Synergistic / Bone-mounted — LoA 1–2

Mazor/MARS is a **bone-mounted (patient-mounted)** mini-robot anchored rigidly to the vertebra. Kinematic structure: **6-DOF parallel mechanism** (Stewart–Gough platform variant). Specifications: $50 \times 50 \times 80\text{ mm}^3$; 250 g; workspace $20 \times 20 \times 20\text{ mm}^3$; drill guide accuracy $< 0.1\text{ mm}$.

Since the robot's reference frame is fixed to the bone, **all global patient movements (breathing, table repositioning) are automatically compensated** — the primary advantage over floor-mounted systems, which require continuous real-time tracking. Clinical context: pedicle screw corridors are **5–7 mm** (lumbar; less thoracic); historical malpositioning rate **10–40%**.

Chapter 4

Neurosurgical Systems

NeuroMate

Origin / Manufacturer: Renishaw Mayfield, UK
Year: 1990s
Domain: Neurosurgery — Stereotactic Guidance
Troccaz / LoA: Semi-active — LoA 1–2

NeuroMate is a **5-DOF** arm for stereotactic neurosurgical procedures (DBS electrode implantation, biopsy). It is the canonical **semi-active** example in Troccaz: the robot positions a mechanical guide tube based on pre-operative imaging, then **locks its motors**. The constraint is **mechanically fixed** once locked; the surgeon manually inserts the tool through the stationary guide.

CyberKnife

Origin / Manufacturer: Accuray
Year: 1999
Domain: Radiosurgery
Troccaz / LoA: Active — LoA 2–3

CyberKnife combines a **KUKA 6-DOF anthropomorphic arm** with a **6 MeV linear accelerator (LINAC)** for **frameless, image-guided stereotactic radiosurgery**. Targeting accuracy: **sub-millimetre**. An anthropomorphic arm is required (not SCARA) because non-isocentric delivery demands radiation beam approaches from arbitrary 3D angles.

With the **Synchrony** respiratory motion tracking system, the arm continuously follows internal tumour position correlated with external breathing markers in real time — elevating effective autonomy to **LoA 2–3**. This is one of the highest-autonomy systems in clinical use.

Chapter 5

Synergistic and Passive Navigation Systems

PADyC

Origin / Manufacturer: TIMC Laboratory, Grenoble
Year: 1990s
Domain: Orthopaedics / MIS
Troccaz / LoA: Synergistic (passive hardware) — LoA 1

PADyC (Passive Arm with Dynamic Constraints) implements virtual fixtures entirely in hardware via opposing **freewheels** on each joint. By controlling freewheel motor speeds, the system defines a window of admissible joint velocities $[\omega_i^-, \omega_i^+]$. The system is **intrinsically safe**: it cannot generate active forces on the patient. Safety constraints are expressed as:

$$WAC(t) = \bigcap_{j=1}^k WAC_j(t)$$

Five operative modes: free, position, region (keep-inside/keep-outside), trajectory, specialised (linear/planar/conical). Main limitation: high computational cost of mapping Cartesian constraints into joint-space WAC; ill-conditioned near singularities.

Viewing Wand (ISG Technologies)

Origin / Manufacturer: ISG Technologies
Year: Early 1990s
Domain: Neurosurgery / Orthopaedics
Troccaz / LoA: Passive — LoA 0

An **articulated passive arm** with joint encoders. The surgeon moves it manually; the system reads joint angles and displays the tip position overlaid on pre-operative CT/MRI. Canonical **passive system** (Troccaz) and **mechanical localiser** (Joskowicz). Advantage over optical trackers: no line-of-sight requirement. Largely superseded by optical tracking today.

Chapter 6

Surface Tracking, Radiology and Ultrasound

SCALPP

Origin / Manufacturer: European research project
Year: 2000s
Domain: Dermatology — Autonomous Skin Harvesting
Troccaz / LoA: Active — LoA 2

A **6-DOF SCARA** robot for autonomous skin graft harvesting. Notable for illustrating two design constraints: (1) the **asepsis rule** — non-medical electronics must remain ≥ 400 mm from the operating table; (2) an **asymmetric wrist** design, motivated by the fact that standard spherical wrist singularities can appear anywhere within the reachable workspace.

OTELO / MELODY

Origin / Manufacturer: French research consortium
Year: 2000s
Domain: Tele-echography
Troccaz / LoA: Teleoperated — LoA 1

An ultra-lightweight robot (**PPRRRP**, 6 DOF) for **remote ultrasound examination**. Its probe-handling end-effector implements an **RCM constraint** to safely maintain skin contact, demonstrating that RCM appears in any task requiring pivoting about a fixed body contact point — not only in laparoscopic MIS.

INNOMOTION

Origin / Manufacturer: Innomedic, Germany
Year: 2000s
Domain: Interventional Radiology — Needle Positioning
Troccaz / LoA: Active / Semi-active — LoA 2

A **5-DOF MR-compatible** system for CT/MRI-guided needle positioning inside the scanner bore (gantry diameter 60–70 cm). MR compatibility mandates non-ferromagnetic materials and pneumatic or piezoelectric actuation, severely constraining design choices.

XACT Robotics

Origin / Manufacturer: XACT Robotics, Israel

Year: 2010s

Domain: Interventional Radiology — CT-guided Procedures

Troccaz / LoA: Active — LoA 2

A **parallel robot** for CT-guided needle-based procedures (biopsy, ablation, drainage). The robot autonomously positions the needle guide to the pre-planned trajectory from the CT scan, operating fully inside the gantry without intraoperative surgeon guidance of insertion angle.

Chapter 7

Cardiac and Microscopy Systems

Cardiolock

Origin / Manufacturer: TIMC Laboratory, Grenoble — research prototype
Year: 2000s
Domain: Cardiac Surgery — Active Motion Compensation
Troccaz / LoA: Cooperative — LoA 1

Cardiolock is an **active cardiac stabiliser** for beating-heart coronary surgery — avoiding extracorporeal circulation (ECC: a pump oxygenating blood while the heart is stopped). Passive stabilisers (Medtronic Octopus, Esculap FlexHeart) leave residual motion of several mm; anastomosis requires ~ 0.1 mm precision.

Architecture: positioning arm + rigid shaft ($L = 250$ mm, trocar-inserted) + stabilising fingers + high-speed camera + **active 2-DOF spherical base** (each DOF is a planar **3-PRR** parallel mechanism).

Actuation: piezoelectric stack actuators; bandwidth hundreds Hz to kHz; stroke **100–200 μ m**; zero backlash; sub-ms response. Classical motors excluded due to gear backlash and insufficient bandwidth for cardiac frequencies (~ 1 Hz with harmonics to 10 Hz).

Stroke amplification via Type 2 singularity: the 3-PRR operates near a parallel singularity where end-effector/actuator velocity ratio diverges:

$$\dot{\theta} = \frac{1}{\|EB_2\| \sin \varepsilon} \dot{q}_2 \quad (\varepsilon \rightarrow 0 \Rightarrow \text{amplification} \rightarrow \infty)$$

Tip kinematics:

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} L \sin \alpha_1 & 0 \\ 0 & L \sin \alpha_2 \end{pmatrix} \begin{pmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{pmatrix}$$

Increasing angles α raises range but increases inertia, degrading dynamic performance (**range/bandwidth trade-off**).

Why not adopted: (1) Safety — uncontrolled motion near the heart upon controller failure; (2) Mechanical bulk; (3) Marginal clinical gain over high-quality passive stabilisers. Later versions replaced rigid joints with compliant mechanisms, analysed via the **Pseudo-Rigid Body Model (PRBM)**.

Pentagon Mechanism

Origin / Manufacturer: TIMC Laboratory, Grenoble — research prototype

Year: 2000s

Domain: In-vivo Microscopy — Physiological Motion Compensation

Troccaz / LoA: Cooperative — LoA 1

A **5-bar closed-loop planar parallel mechanism** driven by two **Physik Instrumente P239.90** piezoelectric actuators (stroke $180\text{ }\mu\text{m}$). Compensates two physiological components:

- **Cardiac:** $20\text{--}40\text{ }\mu\text{m}$ at $\sim 10\text{ Hz}$.
- **Respiratory:** $1000\text{--}1200\text{ }\mu\text{m}$ at $\sim 0.2\text{ Hz}$.

Jacobian computed via the **Virtual Tree Method** (cut one link; impose zero velocity at cut point). Two co-optimised design objectives via Jacobian eigenvalue analysis: (1) **Sufficient amplification** — achieved $563\text{ }\mu\text{m}$ (actuator 1) and $464\text{ }\mu\text{m}$ (actuator 2) from $180\text{ }\mu\text{m}$ stroke; (2) **Isotropy** — condition number minimised for uniform amplification in all directions.

Chapter 8

Ophthalmic Robotic Systems

Steady-Hand Robot (SHR)

Origin / Manufacturer: Johns Hopkins University (JHU)
Year: 2000s
Domain: Ophthalmic Surgery — Retinal Microsurgery
Troccaz / LoA: Cooperative — LoA 1

Clinical motivation: retinal vein diameter $\sim 100 \mu\text{m}$; surgeon RMS tremor $182 \mu\text{m}$ — freehand cannulation is physically infeasible. SHR operates in **cooperative admittance** mode: $v = G f$, $G \in \mathbb{R}^{6 \times 6}$. The RCM is enforced **entirely in software** as an optimisation constraint at each control step — not via a mechanical linkage, making the system compact but software-dependent. Required DOF for retinal surgery: 4 (3 orientation rotations + 1 axial translation).

Micron (Carnegie Mellon University)

Origin / Manufacturer: Carnegie Mellon University (CMU)
Year: 2000s
Domain: Ophthalmic Surgery — Tremor Cancellation
Troccaz / LoA: Cooperative / Handheld — LoA 1

Micron is a **fully handheld active micromanipulator**: unlike every other system in this guide, it shares **no mechanical support with a robotic arm**. The surgeon holds it directly. Embedded **piezoelectric actuators** sense and cancel tremor locally in real time, exploiting their kHz bandwidth to track physiological tremor frequencies.

Leuven Robot

Origin / Manufacturer: University Hospitals Leuven, Belgium
Year: 2000s
Domain: Ophthalmic Surgery — Retinal Microsurgery
Troccaz / LoA: Cooperative — LoA 1

Enforces RCM via a **mechanical parallel linkage** — hardware-guaranteed safety, in contrast to SHR's software approach. Operates in cooperative mode for retinal procedures.

TU Munich Ophthalmic Robot

Origin / Manufacturer: Technical University of Munich

Year: 2000s

Domain: Ophthalmic Surgery

Troccaz / LoA: Cooperative — LoA 1

Hybrid parallel-serial architecture: the parallel sub-structure provides high stiffness and spatial positioning accuracy; the serial wrist provides the orientation dexterity required for intraocular manipulation. Illustrates why hybrid architectures exist: no single topology satisfies all requirements simultaneously.

Chapter 9

Radiotherapy and Application-Specific Systems

ROBHOT / Alba ON4000

Origin / Manufacturer: Sapienza University of Rome / Medlogix s.r.l.

Year: 2010s

Domain: Oncological Hyperthermia

Troccaz / LoA: Cooperative — LoA 1

ROBHOT positions an electromagnetic antenna on the body surface to deliver **oncological hyperthermia**: raising tumour temperature to **40–45°C** (+4–8°C above baseline). Biological premise: tumour cells are more thermosensitive than healthy tissue; heat inhibits DNA repair mechanisms, enhancing radio/chemotherapy. Uses two control laws simultaneously: **impedance control** for autonomous antenna path tracking; **admittance control** for manual operator repositioning.

Registration chain:

$${}^A p_T = {}^A T_{A_m} {}^{A_m} T_V {}^V T_{CT} {}^{CT} p_T$$

where ${}^V T_{CT}$ is the ICP Gauss-Newton registration output (constant); ${}^{A_m} T_V$ is measured live by the **NDI Polaris Vega** optical camera; ${}^A T_{A_m}$ is fixed calibration. Validation RMS error: <1 cm.

Surgiscope

Origin / Manufacturer: ISIS / Elekta

Year: 1990s–2000s

Domain: Ophthalmic and Neurosurgery

Troccaz / LoA: Hybrid — LoA 1

A **hybrid** robotic microscope support combining a parallel positioning sub-structure with a serial wrist. Cited in the Dario (2004) classification as an example of hybrid architecture — parallel for stiffness, serial for orientation dexterity.

Chapter 10

Quick Reference Table

System	Domain	Troccaz	LoA	Key distinguishing feature
PUMA 560	Neurosurgery	Active	2	First surgical robot (1985)
ROBODOC	Orthopaedics	Active	2	SCARA; Orthodoc; <1 mm milling
CASPAR	Orthopaedics	Active	2	Anthropomorphic; real-time fiducial set S'
ZEUS	MIS / Telesurgery	Teleoperated	1	Operation Lindbergh; PRRRR passive RCM
da Vinci	MIS	Teleoperated	1	7 DOF EndoWrist; $10 \times 3D$; 3:1/5:1 scaling
AESOP	MIS	Active (Aux.)	2	Voice-controlled 7-DOF camera arm
MAKO	Orthopaedics	Synergistic	1	3-region impedance; 550/100 Hz dual loop
Mazor/MARS	Spinal surgery	Synergistic	1–2	Bone-mounted Stewart; <0.1 mm; auto patient-motion comp.
NeuroMate	Neurosurgery	Semi-active	1–2	5-DOF; locked mechanical guide
CyberKnife	Radiosurgery	Active	2–3	KUKA 6-DOF + LINAC 6 MeV; sub-mm
PADyC	Orthopaedics	Synergistic	1	Freewheel passive VF; intrinsically safe
Viewing Wand	Neuro/Orthopaedics	Passive	0	Passive arm navigator; no line-of-sight issue
SCALPP	Dermatology	Active	2	SCARA; 400 mm asepsis rule
OTELLO/MELODY	Tele-echography	Teleoperated	1	PPRRRP; remote probe; RCM on skin
INNOMOTION	Interv. radiology	Active	2	5-DOF; MR-compatible
Cardilock	Cardiac surgery	Cooperative	1	3-PRR singularity amplification; piezo
Pentagon	In-vivo microscopy	Cooperative	1	5-bar; $563 \mu m$ amplification; isotropy
SHR	Ophthalmology	Cooperative	1	Software RCM; admittance $v = Gf$
Micron	Ophthalmology	Cooperative	1	Handheld; embedded piezo tremor cancellation
Leuven Robot	Ophthalmology	Cooperative	1	Mechanical RCM via parallel linkage
TU Munich	Ophthalmology	Cooperative	1	Hybrid parallel-serial
ROBHOT	Hyperthermia	Cooperative	1	Impedance+admittance; NDI Polaris Vega
Surgiscope	Ophthalmic/Neuro	Hybrid	1	Hybrid; microscope stabilisation
XACT Robotics	Interv. radiology	Active	2	Parallel; CT-guided needle placement

Table 10.1: All surgical robots covered in the course.